

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.1 TAXONOMY I: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Your task is to answer the driving question: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? You will do two things. First, with your group, make a local-organism classification guide or poster using organisms such as a goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito. Second, write your own explanation. Use the model answers below to see what a strong explanation looks like. Each section has a prompt and an exemplar. Write in your own words. Use evidence from our lessons, such as the sorting activity, the hierarchy of taxonomic units, binomial names and the five kingdoms. Always connect your ideas back to the puzzle we started with: everyone sorted the organisms differently and used different local names, yet each person was right by their own rules.

1. Why do we need a classification system?

Describe what happened when your group sorted the organisms and compared with other groups. Explain why the different groupings and different local names created a problem, and why a shared system is needed.

When we sorted the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito, my group sorted by usefulness (food, pests, disease). Another group sorted by colour and another by where the organisms live. Each group was correct by its own rule, but we ended up with different groups for the same organisms. Names made it harder: the same organism can have different names in Kikuyu, Luo, Kiswahili and English, and one name can mean different things in different places. There are also millions of kinds of living things, so it is very hard to study or identify them one by one. This is why we need taxonomy, a scientific system for identifying, naming and grouping organisms by their similarities and characteristics, which everyone in the world uses in the same way.

2. How are organisms arranged in groups? (The hierarchy)

Explain how taxonomic units are arranged from Kingdom to Species. Use one organism from your poster, such as maize, to show how the levels become more specific.

Scientists arrange organisms in a hierarchy of taxonomic units: Kingdom, Phylum (Division in plants), Class, Order, Family, Genus and Species. A kingdom is the biggest group and contains a huge variety of organisms. As you move down each level, the groups become smaller and the organisms in them share more and more characteristics. For example, maize belongs to the Kingdom Plantae, which also includes bean plants and sukuma wiki. Moving down the levels, fewer and fewer organisms remain in each group until we reach the species level, where organisms are so similar that they can breed with each other. Maize and a bean plant share the kingdom but are in different lower groups because they differ in other characteristics. This step-by-step arrangement

	means a scientist in Kisumu and one in Nairobi can place an organism in exactly the same position.
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3. How do scientists name organisms? (Binomial nomenclature)

Explain what binomial nomenclature is, how a scientific name is written, and why it solves the problem of different local names.	Binomial nomenclature is the system of giving every organism a two-part scientific name made up of the genus and the species. For example, maize is <i>Zea mays</i> . <i>Zea</i> is the genus (it starts with a capital letter) and <i>mays</i> is the species (it starts with a small letter). The name is written in italics, or underlined if you are writing by hand. This name is the same in every country and every language. So whether someone calls maize 'mahindi' in Kiswahili or 'maize' in English, scientists everywhere know that <i>Zea mays</i> means the same plant. This solves the problem we saw in class where one organism had many local names and no one could be sure they were talking about the same thing.
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4. What characteristics place organisms into the five kingdoms?

Using cell structure, body organisation and mode of nutrition, explain how the organisms on your poster fit into the five kingdoms: Monera, Protocista, Fungi, Plantae and Animalia.	Scientists use characteristics such as cell structure (whether the cell has a nucleus), organisation (single-celled or many-celled) and mode of nutrition (how it gets food) to place organisms in the five kingdoms. Bacteria are in Monera because they are single-celled and have no true nucleus. Algae are in Protocista because they are mostly single-celled or simple organisms with a nucleus; many make their own food by photosynthesis. A mushroom is in Fungi because it has a nucleus and feeds by absorbing nutrients from dead or living matter, not by photosynthesis. Maize and bean plants are in Plantae because they are many-celled and make their own food using chlorophyll. A goat, a chicken and a mosquito are in Animalia because they are many-celled and take in food made by other organisms. This shows that the groups we make must be based on shared characteristics that are reliable, not on colour or usefulness alone.
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5. Why does taxonomy matter for people and conservation?

Put your whole explanation together. Answer the driving question and explain how taxonomy helps people in Kenya to identify, study, communicate about and conserve living organisms.	We can organise the many living organisms around us by using taxonomy: grouping them in a hierarchy from Kingdom to Species based on shared characteristics, and giving each one a universal two-part scientific name. Because of this, a farmer in Kericho, a student in Mombasa and a scientist abroad can identify and study the same organism in the same way, even when their local names are different. For instance, knowing that a mosquito is an animal, and which kind it is, helps health workers around Lake Victoria study and control malaria. Knowing the correct species of plants and animals also helps conservationists protect them, for example the wildlife in the Rift Valley, because you cannot protect what you cannot correctly identify and name. So our class sorting problem has been solved: instead of many different personal systems, taxonomy gives one shared system for identifying, studying, communicating about and conserving the diversity of life.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of characteristics (30%)	Consistently uses cell structure, body organisation and mode of nutrition correctly to place every organism in the right kingdom, and explains why other	Uses most characteristics correctly to place organisms in kingdoms, with minor gaps or one or two errors in explanation.	Uses few characteristics or relies mainly on colour, size or usefulness; several organisms are placed with weak or missing reasons.

	features such as colour or usefulness are not reliable.		
Evidence and justification (25%)	Supports every claim with specific evidence from the lessons (the sorting activity, hierarchy, binomial names, kingdom characteristics) and links reasoning clearly to the driving question.	Supports most claims with evidence from the lessons; the link to the driving question is present but not always clear.	Makes claims with little or no evidence; reasoning is unclear or does not connect to the driving question.
Accuracy of classification and naming (20%)	Hierarchy is in the correct order and binomial names are written correctly (genus capitalised, species lower case, italicised or underlined) with correct kingdoms.	Hierarchy and names are mostly correct, with one or two small errors in order, format or kingdom.	Several errors in hierarchy order, name format or kingdom placement.
Communication and collaboration (15%)	Poster and written explanation are clear, well organised and use scientific terms correctly; the student contributed fully and respectfully to the group work.	Work is mostly clear and organised with some scientific terms; the student contributed to the group work.	Work is hard to follow, scientific terms are missing or misused; there is limited contribution to group work.
Local application and conservation connection (10%)	Uses several Kenyan examples and clearly explains how taxonomy supports identification, communication and conservation of local biodiversity.	Uses some Kenyan examples and mentions a link to conservation.	Few or no local examples; conservation link is missing or unclear.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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